

Integrating trust and satisfaction into the UTAUT model to predict Chatbot adoption – A comparison between Gen-Z and Millennials[☆]

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ABSTRACT

This paper examines the key determinants of behavioral intention, user satisfaction, and chatbot adoption among urban, college-educated student populations within Generation Z and Millennials in India. While Millennials grew up with the Internet, Gen Z was born into the era dominated by social media and smartphones, making them inherently tech-savvy and drawn to chatbots for information access. This study extends the Unified Theory of Acceptance and Use of Technology (UTAUT) by integrating technological elements with trust and satisfaction to propose a conceptual model. Using a mixed-method approach, data were collected through a cross-sectional online survey of 487 chatbot users from urban educational institutions in India. Partial Least Squares Structural Equation Modeling (PLS-SEM) was used to test 11 hypothesized direct relationships. The results suggest that users' willingness to adopt chatbots is significantly influenced by performance expectancy, social influence, trust, and satisfaction. Regarding user satisfaction, both facilitating conditions and trust played substantial roles. Additionally, this study found meaningful associations between facilitating conditions, satisfaction, intention, and adoption. Multi-group analyses revealed notable differences in chatbot adoption factors between Gen Z and Millennials within the study's sampled population. Given the importance of trust in chatbot adoption, the paper highlights that reducing perceived risks can strengthen trust, enhance user satisfaction, and drive chatbot intention and adoption. The above findings offer context-specific insights for chatbot providers in devising strategies to improve user trust, satisfaction, and adoption within similar demographics.

1. Introduction

In recent years, organizations have offered many artificial intelligence (AI) tools to enhance customer service and support. The promise of superior customer experience through quick response and 24/7 support has increased the use of chatbots. Besides automating significant areas of customer service operations, it also augments the corporate sales force (Hildebrand & Bergner, 2019). A 2024 report by Mordor Intelligence estimated the global chatbot market at USD 7.01 billion in 2024 and forecasted to grow at a compound annual growth rate (CAGR) of 24.32 % from 2024 to 2029 (Mordor Intelligence, 2024). Data show that the USA, India, Germany, the UK, and Brazil are the leading countries in chatbot usage. Over the past five years, interest in chatbots within these countries has surged fivefold, and market predictions suggest a valuation of USD 1.3 billion by 2025 (AIMultiple, 2023). As chatbot technology becomes more mainstream, its adoption across

industries is set to expand further.

The Asia Pacific market, led by economies like China, India, Indonesia, Vietnam, etc., experienced exponential growth in chatbot adoption due to accelerated retail and e-commerce business. However, as per industry reports, consumer trust in chatbots is variable. As per TechSee (2022) State of the Service Market Report, only 1 in 3 customers trust chatbots to provide satisfactory service. This varying level of trust has a detrimental effect on adoption rates and overall satisfaction with the tool. As per a report from the Institute of Customer Service, if chatbots are not well designed, it can affect customers' trust and satisfaction in using chatbots overall (Verdict, 2022). As per the findings, two in five people avoid chatbots while trying to get answers to complex questions. While most respondents perceived chatbots as easy to connect, around 30 % felt that chatbots could successfully assist them while only 10 % perceived that chatbots understood them well (Userlike, 2022).

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A survey by [Tidio \(2023\)](#) found that 53 % of respondents cited long waiting times with agents as highly frustrating, resulting in 62 % favoring chatbots over waiting for customer service representatives. However, contrasting insights from [Userlike \(2022\)](#) show that 60 % of the respondents would rather wait in a queue to interact with a human agent than immediately connect with a chatbot. While appreciating the speed, convenience, and all-time availability of chatbots, the respondents reported negative aspects, such as inability to understand and resolve the query, confusion, and unfriendliness. Despite these potential downsides, its mass adoption in personal and professional lives indicates its usefulness ([Capgemini, 2023](#)).

In a systematic literature review, [Jenneboer et al. \(2022\)](#) examined factors driving customer loyalty toward chatbots. They identified customer trust and satisfaction as the most frequently discussed across the reviewed studies. To date, relatively little research has been conducted applying UTAUT and dimensions of trust and satisfaction to examine individuals' decision to use chatbot applications. The main objective of this study is to examine the factors that impact user satisfaction, intention, and adoption, using the UTAUT model as its foundational framework. Furthermore, this study incorporates trust as an extended variable into UTAUT to address the security experience of using chatbots. This focus on trust is anchored on the premise that, in online interactions where rewards are uncertain, trust becomes a critical factor that shapes the level of satisfaction within the relationship ([Montazemi & Qahri-Saremi, 2015](#)).

Prior studies have shown that Millennials or Generation Y (age: 27–43 years; born: between 1981 and 1996) and Generation Z (age: 11–26 years; born: between 1997 and 2012) are highly keen on embracing emerging technologies including chatbots for instant gratification. As per a study by [Userlike \(2022\)](#), about 75 % of respondents above 45 had never used a chatbot, and most users either belonged to the Gen Z or Millennial cohort. Gen Z and millennials are most willing and likely to try out a chatbot. [Forbes \(2018\)](#) reports that 60 % of millennials have already interacted with chatbots, while the remaining 40

% express interest in trying them. On the contrary, while the youngest cohort, Gen Z spends more time online, they are skeptical of chatbots with only 44 % of the opinion that chatbots will make traditional customer support call centers obsolete ([Tidio, 2023](#)). Thus, there is a gap concerning understanding how millennials and Gen Z differ in their involvement with chatbots.

In addition to the numerous benefits that chatbots provide, this study addresses two key research questions:

RQ1: Can chatbots improve customer satisfaction and strengthen user intention and adoption?

RQ2: Do generational characteristics, particularly age differences among Millennials and Gen Z, influence the relationship between factors leading to chatbot intention and adoption?

The research objectives are twofold: first, to provide a greater understanding of the important factors influencing chatbot adoption among Gen Z and Millennials, and second, to provide theoretical implications regarding UTAUT and practical implications for organizations providing customer support chatbots. The section below overviews chatbots' expanding role in customer service and support. It then explores two key models that explain user adoption and satisfaction: the UTAUT model and the cognitive model of satisfaction. The third section outlines the methodology to examine the proposed hypothesis and the conceptual model. This is succeeded by an outline of the study's findings and an analysis of the results. The second-to-last section outlines both scholarly and marketing implications, while the concluding section examines the study's limitations and proposes avenues for future studies. A snapshot of the flow is depicted in [Fig. 1](#).

2. Literature review

In this section, we explore using chatbots as conversational agents, explore relevant technology adoption models, and examine the satisfaction model. Additionally, it formulates hypotheses based on constructs and relationships drawn from technology adoption theories,

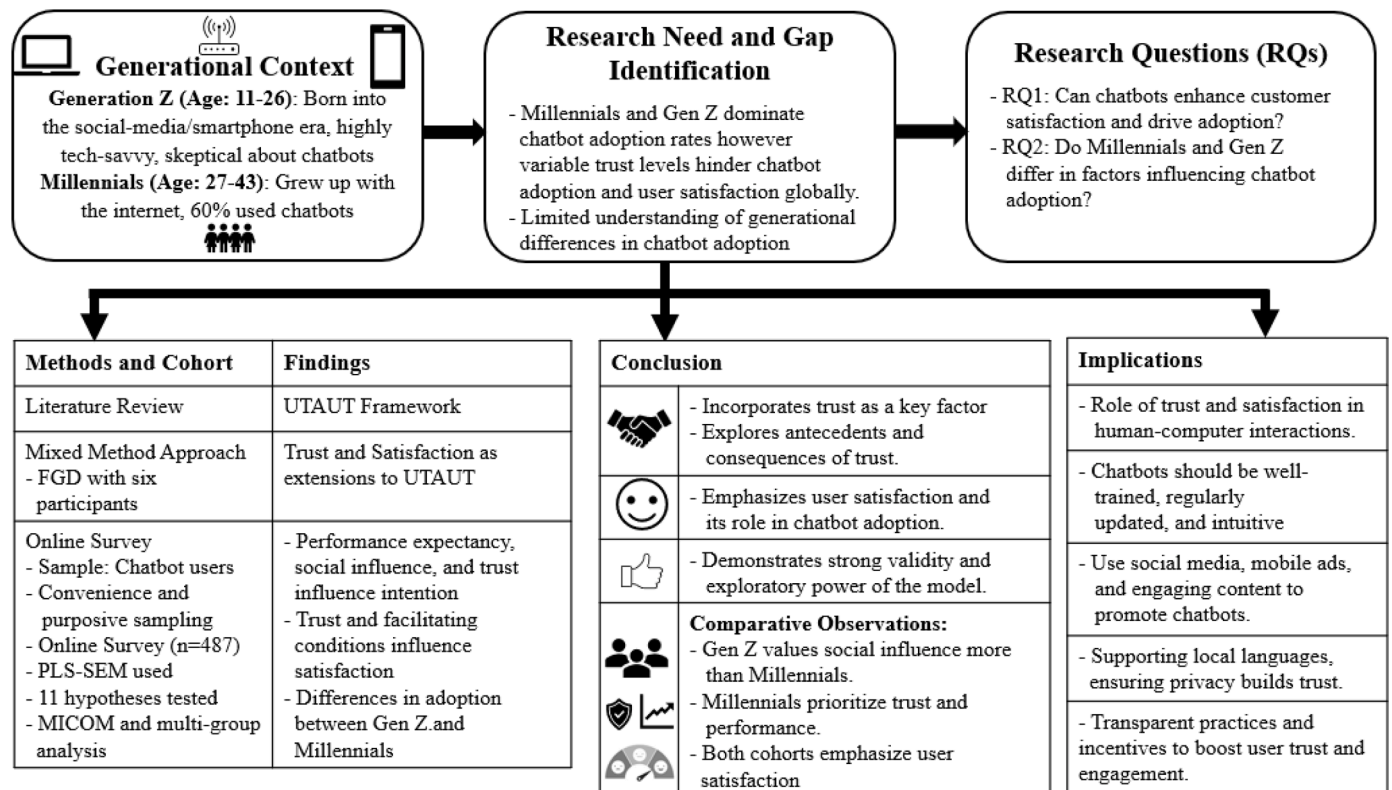


Fig. 1. Research flow - determinants of chatbot adoption among Gen Z and Millennials.

expecting that the general causal relationships identified in these models will apply to chatbots.

2.1. Chatbots in conversational roles

The word "chatbot" combines "chat," which refers to conversation, and "bot," which is an abbreviation for robot or automated system—essentially referring to an application that interacts with people using artificial intelligence (Rese et al., 2020). Today's chatbots are designed to mimic human conversational behaviors (Murtarelli et al., 2023) and are poised to revolutionize human interaction in all possible fields (Kasilingam, 2020).

Prior studies in chatbot adoption involving generational cohorts have mainly used the theoretical lens of either the TAM or the UTAUT (Table 1). Consequently, this study extends the UTAUT model with satisfaction to examine the drive and intention to use chatbots. Most recent studies have identified key factors for chatbot adoption, with attitude, intention, and adoption as the dependent variables. Thus, customer satisfaction, a significant reason for low acceptance rates, has not been considered. In addition, few research studies (Table 1) have attempted to study the adoption differences between cohorts, e.g., Gen Z and millennials in this study. Recent systematic literature reviews involving the comprehensive analysis of 83 studies (Rapp et al., 2021), 54 studies (Li et al., 2023), and 40 articles (Ng & Zhang, 2025) emphasize the importance of ease of use, perceived usefulness, trust, and user satisfaction as key determinants of chatbot adoption.

2.2. Significant frameworks in technology adoption studies

2.2.1. UTAUT

Introduced by Venkatesh et al. (2003), the Unified Theory of Acceptance and Use of Technology (UTAUT) is considered among the predominantly referenced frameworks that clarify the factors driving technology adoption and use. According to UTAUT, factors that influence behavioral intention are performance expectancy (PE), effort expectancy (EE), social influence (SI), and facilitating conditions (FC). EE emphasizes simplicity and ease of use in this framework, while PE focuses on improving business outcomes. Later, constructs namely hedonic motivation, price value, and habit were added as an extension to UTAUT (Venkatesh et al., 2012).

Previous researchers have incorporated constructs like trust and satisfaction into TAM (Choung et al., 2023; Murtarelli et al., 2023). Shin (2010) extended the UTAUT framework by including additional elements like trust, security, social influence, and self-efficacy. In a recent study in the Indian context, Menon and Shilpa (2023) extended the UTAUT with two constructs - perceived interactivity and privacy - to explain ChatGPT intention. Numerous studies have confirmed that UTAUT is reliable for understanding technology adoption. To explain user acceptance of service robots, the Service Robot Acceptance Model (SRAM), which consists of relational (trust and privacy) and socio-emotional (functional) components, was later proposed as an extension to the TAM and UTAUT models (Wirtz et al., 2018). However, this model ignores user satisfaction, which is important and primarily motivated by expectations being met. This limitation therefore necessitates adding trust and satisfaction aspects to the UTAUT model, which should provide a strong foundation for examining chatbot adoption. In contrast to other models, UTAUT's comprehensive approach and good explanatory power are the reasons for choosing it (Venkatesh et al., 2003).

Additionally, UTAUT has been customized for several applications, such as chatbots in a variety of industries like banking (Trivedi, 2019; Hari et al., 2022), education (Smutny & Schreiberova, 2020); Chocarro et al., 2021), healthcare (Palanica, 2019), tourism (Pillai & Sivathanu, 2020; Melián-González et al., 2021), online shopping (Kasilingam, 2020) and customer service (Nordheim et al., 2019; Adam et al., 2021).

Table 1

Recent studies on chatbot adoption by Generation Z and Millennials.

Authors	Focus, sample & geography	Theoretical lens	Key findings
Arce-Urriza et al. (2025)	Role of GenAI on consumer adoption of retail chatbots (United States)	Service Robot Acceptance Model (SRAM)	Chatbot usefulness, human-likeness, and familiarity, enhances adoption intentions. GenAI does not alter trust but increases privacy concerns.
Bhatnagar and Rajesh (2024)	Role of artificial intelligence characteristics on Gen Y and Gen Z user satisfaction and intention towards digital banking apps (India)	Expectancy Confirmation Theory (ECT)	Perceived intelligence and animacy influence anthropomorphic perceptions, enhancing usefulness, satisfaction, and intention.
Sharma et al. (2024)	Role of digital assistants' attributes on Millennials purchasing intention (India)	Uncanny Valley Theory (UVT)	Perceived interactivity, anthropomorphism, and social presence positively influence purchase intention with gender moderating the relationship between perceived interactivity and purchase intention.
ElSayad and Mamdouh (2024)	Examine the impact of technology readiness motivators and inhibitors on perceived trust, usefulness, and purchase intention towards AI-powered retail platforms among Millennials and Gen Z (Egypt)	Technology Readiness and Adoption Model (TRAM)	Perceived trust and usefulness significantly influence purchase intention, with trust mediating the relationship between motivators and intention. Optimism also significantly influences trust and usefulness.
Murtarelli et al. (2023)	Identifying factors affecting the quality of millennials-Chatbot Relationships (Italy)	Technology Acceptance Model (TAM)	Perceived ease of use (PEOU) and perceived usefulness (PU) positively influence the attitude; perceived trust, perceived risk, and attitude influence behavioral intention.
Choung et al. (2023)	Role of trust in users' intention to adopt AI technologies	Technology Acceptance Model (TAM)	Perceived usefulness, ease of use, trust, and attitude, significantly influence usage intention.
Dinh and Park (2023)	An empirical analysis of how utilitarian and hedonic motivation along with social presence influence intention and examining the moderating effect of generational cohorts (Baby boomers and Generations X, Y, Z) (Taiwan)	Self-determination theory (SDT)	Hedonic but not utilitarian motivation significantly affects chatbots' social presence, ultimately influencing the intention to use the chatbot. Moderation effect of generational cohorts established.
Maar et al. (2023)	Investigating how chatbot, customer, and contextual characteristics, chatbot warmth and competence, shape customers' chatbot-related attitudes of	Stereotype Content Model (SCM)	Gen Z shows more positive attitudes toward chatbots than Gen X

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Table 1 (continued)

Authors	Focus, sample & geography	Theoretical lens	Key findings
Toh and Tay (2022)	GenX and Gen Z users (Germany) Examine the acceptance of Banking Chatbots among millennials (Malaysia)	Unified Theory of Acceptance and Use of Technology (UTAUT)	Performance expectancy (PE), effort expectancy (EE), facilitating conditions (FC), social influence (SI), hedonic motivation, habit, and compatibility explain the acceptance of chatbots
Ameen et al. (2022)	Examine how augmented reality, artificial intelligence chatbots, and social media impact the image and self-esteem of Generation Z female consumers (Malaysia)	Social Comparison Theory	AI-enabled chatbot experiences and trust affect the image, self-esteem, and purchase behavior.
Joshi (2021)	Adoption of Customer Service Chatbots among Millennials (India)	UTAUT and Cognitive Model of Satisfaction	PE and SI significantly impact intention whereas Trust and FC influence satisfaction.
De Cicco et al. (2020)	Millennials' attitude toward chatbots (Italy)	Social Presence Theory	Social presence positively influences trust and perceived enjoyment, which positively influences attitude.
Rese et al. (2020)	To measure the acceptance of chatbots among Millennials (Germany)	Technology Acceptance Model (TAM) and Uses and Gratification (U&G) Theory	Utilitarian factors (authenticity of conversation, PU) and hedonic factors (perceived enjoyment) positively influence acceptance

2.2.2. A cognitive model of satisfaction decisions

Oliver (1980) put forth a framework that uses the interplay between expectancies and expectancy disconfirmation to explain consumer satisfaction. Satisfaction is the difference between users' expectations and actual experiences. According to Liao et al. (2009), an information system's unique characteristics affect users' expectations for results, leading to positive or negative reactions and eventually impacting user acceptance. Both pre- and post-use experience play a role in determining satisfaction or dissatisfaction, which in turn is believed to affect changes in attitudes and intentions to make purchases.

Prior authors (Bargas-Avila et al., 2009; Borsci et al., 2023) have proposed scales to measure satisfaction with online services. Bargas-Avila et al. (2009) conducted a study to develop and validate a questionnaire to measure user satisfaction with the Intranet. Borsci et al. (2023) developed a standardized scale "Bot Usability Scale (BUS)" to assess and compare the satisfaction of chatbot users. It is seen that both scales capture the relevant attributes of satisfaction, such as being easy to understand, clear, personalized, complete, relevant, reliable, private and secure, flexible, integrated, and timely.

Gatzioifa and Saprikis (2022) provided an extensive literature review on chatbot adoption, highlighting that UTAUT was the most commonly used model. Additionally, 51.3 percent of the reviewed studies did not adhere to a single theoretical framework; instead, they created their models by blending aspects from various theories.

2.3. Theoretical framework and hypotheses

2.3.1. Effort expectancy (EE)

Effort Expectancy is "the ease with which a system can be used and operated" (Venkatesh et al., 2003). Hew et al. (2015) state that user-friendly applications attract consumers, influencing their propensity to utilize them. Effort Expectancy (EE) has been found to influence behavior in a variety of technologies, including chatbots (Arce-Urriza et al., 2025), digital wallets (Chawla & Joshi, 2019; Singh et al., 2020), online banking (Suh & Han, 2003), and mobile banking (Deb & David, 2014; Chawla & Joshi, 2018).

Araujo and Casais (2020) evaluated Portuguese users' acceptance of chatbots for shopping using the TAM model. The findings imply that consumers' opinions of chatbots are significantly influenced by Perceived Ease of Use (PEOU), which influences their behavioral intention to utilize them. The PEOU factor aligns with Effort Expectancy, representing the anticipated ease of interaction with a chatbot.

Previous research in industries like higher education (Chatterjee & Bhattacharjee, 2020), hospitality and tourism (Pillai & Sivathanu, 2020), and online shopping (Kasilingam, 2020) has shown that EE is essential for encouraging chatbot adoption across such sectors. More recent studies like those by Nguyen et al. (2021) and by Seo and Lee (2021), have also explored a related concept, "system quality," emphasizing elements such as ease of use, reliability, availability, and response speed in chatbot platforms. Given this literature, the hypothesis we propose is:

H1. Effort Expectancy positively influences the willingness to utilize chatbots.

2.3.2. Performance expectancy

Venkatesh et al. (2003) describe Performance Expectancy as "the extent to which a person believes that using the system will support improvements in job performance". Nguyen et al. (2021) established that when users perceive value in chatbot services for tasks like retrieving information, completing online transactions, receiving swift responses, and achieving practical solutions, it enhances their experience and motivates continued use. Gansser and Reich (2021) utilized the UTAUT2 framework in their research, focusing on three user groups in Germany— household, mobility, and health. Their findings revealed that performance expectancy considerably impacted behavioral intentions and usage behaviors concerning artificial intelligence technologies. Considering this framework, the proposed hypothesis is:

H2. Performance Expectancy positively impacts the intention to utilize chatbots.

2.3.3. Social influence

Ventakesh et al. (2003) defined Social Influence as "the extent to which an individual perceives that important others expect them to use the new system". According to the Theory of Reasoned Action (TRA) (Fishbein & Ajzen, 1975), a person's motivation to adopt a behavior is shaped by social attitudes, often influenced by those in their immediate circle. In technology uptake, this idea is mentioned as subjective norms, where individuals consider how much their peers (e.g., friends, colleagues, supervisors) impact their decision to adopt new technology (Taylor & Todd, 1995). These norms create informal social expectations, encouraging individuals to conform. More substantial social influence from one's peer group is believed to increase the likelihood of adopting technology. From this perspective, the hypothesis proposed is:

H3. Social Influence positively impacts the intention to use chatbots.

2.3.4. Facilitating conditions

The term "Facilitating Conditions" refers to a user's belief that adequate organizational and technical resources are available to simplify system use. This includes external variables that affect a user's

motivation to participate (Teo et al., 2007) and make tasks easier to complete (Thomson et al., 1991). Facilitating conditions include perceptions of available training, support, infrastructure, and prior knowledge necessary for effective system use (Garone et al., 2019). In 2020, Kasilingam observed that users are more likely to embrace smartphone chatbots when necessary technical infrastructure is present.

In technology contexts, organizational and technical support facilitate system adoption (Agarwal et al., 2009). Brachten et al. (2021) observed that the availability of supportive conditions for enterprise chatbots reduces perceived effort, making users more inclined to adopt the technology. Previous research (Lin, 2011; Shaw, 2014) suggests that individual skills, affordable internet access, smartphones, and supportive legal frameworks significantly influence users' adoption intentions. Chatterjee and Bhattacharjee (2020) further indicate that robust technical infrastructure and relevant training positively impact the acceptance of new technologies. Based on these insights, the hypotheses below are proposed:

H4. *Facilitating Conditions positively affect the behavioral intention to use chatbots.*

H5. *Facilitating Conditions positively affect the adoption of chatbots.*

H6. *Facilitating Conditions positively affect satisfaction with chatbots.*

2.3.5. Trust

Trust is "the belief that others will, as much as possible, act in our best interests and refrain from exploiting us" (Baier, 1986). In the technological environment, as per McKnight et al. (2011), trust refers to "the belief that a particular technology possesses the necessary qualities to operate as anticipated in a specific environment" (Joshi, 2021). Caceres and Paparoidamis (2007) suggest that for chatbots, trust is the degree of confidence users have in the system's reliability and quality.

Trust includes two aspects: cognitive and emotional. Cognitive trust is the user's belief that the chatbot provider will exhibit competence, goodwill, and ethical behavior. In contrast, emotional trust pertains to the user's sense of security and comfort with the provider.

From the customer service perspective, the importance of trust originates from the commitment trust theory, which posits that trust engenders commitment in a provider-consumer relationship that leads to greater satisfaction. Applied to the consumer adoption of customer-service chatbots, commitment trust theory (Morgan & Hunt, 1994) suggests that consumers with higher trust are more likely to adopt such services, i.e., in other words, trust in technology is a key determinant of satisfaction, intention, and adoption. Trust is incorporated into the UTAUT in several ways. Previous studies have identified initial trust as a precursor to behavioral intention, for instance, in a study conducted in Lebanon, Mostafa and Kasamani (2022) found that consumers' aim to use chatbots broadly depends on their initial level of trust in these systems. Prior studies on chatbots have conceptualized trust as a key determinant of satisfaction (Chung et al., 2020; Joshi, 2021) leading to higher intention (Phaosathianphan & Leelasantitham, 2019; Kasilingam, 2020; Chatterjee & Bhattacharjee, 2020; Bhatnagr & Rajesh, 2024) and adoption (Følstad et al., 2018).

Letheren and Dootson (2017) suggest that the balance between security and convenience hinges on trust, encompassing the service provider's commitment to protecting personal data and trust in the platform's reliability. Therefore, trust is crucial in realizing the added value of chatbots. Through field and lab research, Hildebrand and Bergner (2019) demonstrated that incorporating chatbots into sales and service operations strengthens consumer relationships, boosts trust, and opens up opportunities for upselling.

This study assesses consumer trust by examining chatbots' competence, goodwill, and ethical standards. Eren (2021) in research with Turkish banking customers using chatbots, discovered that perceived trust in chatbots substantially impacts customer satisfaction. Although

trust has been a focal point in studies on other online services like banking, mobile platforms, and social networks, its exploration in chatbot adoption research is still limited. This study integrates trust within the UTAUT model, identifying it as a critical factor in user satisfaction. Consequently, the below-mentioned hypotheses are proposed:

H7. *Trust positively influences the intention to use chatbots.*

H8. *Trust positively influences user satisfaction with chatbots.*

2.3.6. User satisfaction

Nguyen et al. (2021) propose that when chatbot services meet users' expectations, resulting in satisfaction, it influences their intentions and promotes ongoing use in the longer term. Similarly, Eren (2021) discovered that fulfilling user expectations through chatbots significantly positively affects satisfaction. In related research on chatbots for luxury brands, Chung et al. (2020) identified that communication accuracy, credibility, and competence play essential roles in enhancing satisfaction. Bhatnagr and Rajesh (2024) extend the traditional view of customer satisfaction to digital natives (Gen Y and Z), showing that satisfaction with digital platforms determines immediate and continued usage intentions. Based on these findings, the hypotheses below are proposed:

H9. *Satisfaction positively influences the intention to use chatbots.*

H10. *Satisfaction positively influences chatbot adoption.*

2.3.7. Behavioural intention

Fishbein and Ajzen (1975) described Behavioral Intention (BI) as "a user's perceived likelihood of performing a specific behavior". A firm intention makes the behavior more likely, positioning BI as a key predictor of technology usage patterns. Prior research by Ventakesh et al. (2003), as well as Tarhini et al. (2016), has highlighted BI's substantial role in influencing actual use or adoption in studies on technology acceptance. Actual usage or adoption is when a user begins interacting with the technology, signifying their initial acceptance. Based on this concept, the hypothesis below is proposed:

H11. *Behavioral Intention positively influences chatbot adoption.*

2.3.8. Age as a moderator

Research indicates substantial differences in beliefs, values, attitudes, and behaviors among age groups (Tang & Austin, 2009). Based on this, it is hypothesized that age groups (e.g., Gen Z and Millennials) may influence the connections between various antecedents and outcome variables, as outlined in H1 to H11.

H12. *The age cohort moderates the relationships proposed in H1 to H11.*

Several hypotheses were developed following the literature review. Fig. 2 presents the recommended conceptual model along with associated research hypotheses. This model includes six primary factors from the UTAUT framework: i) performance expectancy, ii) effort expectancy, iii) social influence, iv) facilitating conditions, v) behavioral intention, and vi) adoption. This model also incorporates two chatbot-specific elements – trust and satisfaction –. It outlines the relationships between various determinants and behavioral outcomes such as adoption, intention, and satisfaction.

2.3.9. Control variables

As per Leong et al. (2013), the inclusion of control variables can influence the outcome variable (e.g., behavioural intention, adoption) and have the potential to confound the influence of determinants on technology adoption. Controlling for variables also makes it easier to compare the findings across different demographics. In this study, gender, income, profession, marital status, and city are chosen as control

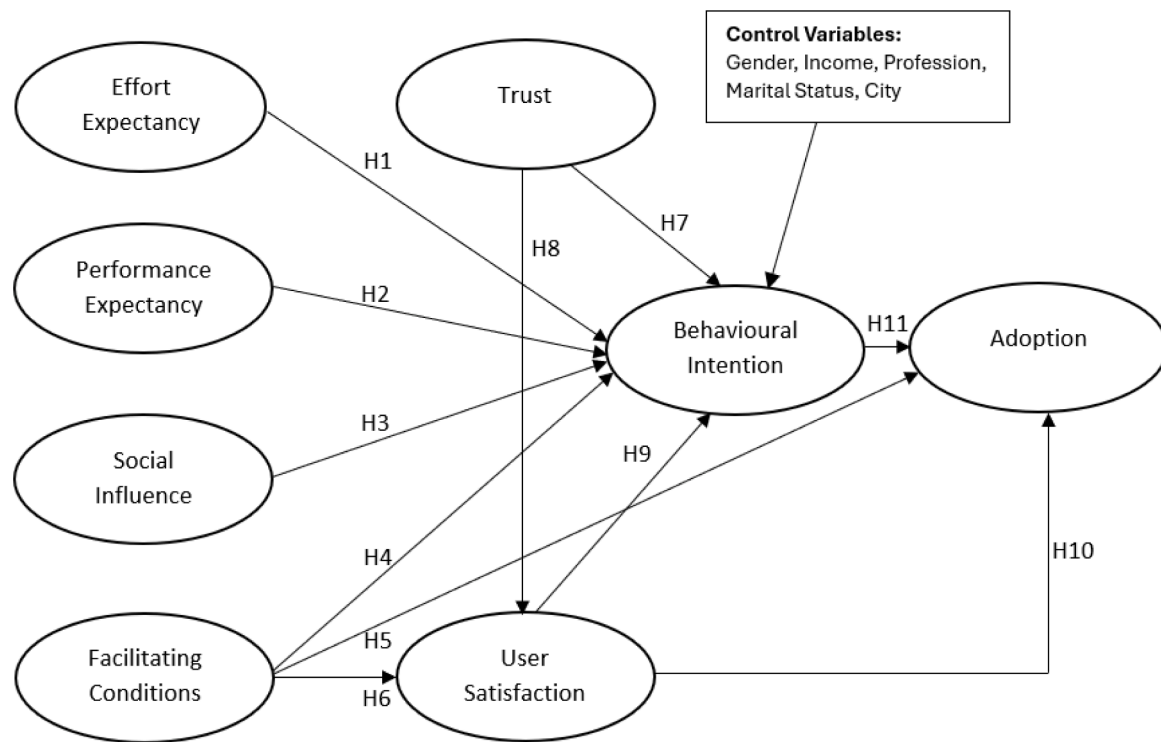


Fig. 2. Conceptual model.

variables.

3. Methodology

The study utilized a mixed-method research design, which included one FGD and an online survey. The comprehensive literature review and FGD insights were used to identify the attributes influencing user satisfaction, intention, and adoption of customer service chatbots. Integrating quantitative and qualitative methods serves two purposes: mutual validation of data and findings (Kelle, 2006) and creating a more coherent and deeper exploration of complex human and social dynamics (Malina et al., 2011.) The chatbot examined was for customer service, like frequently asked questions, placing and tracking orders, providing feedback, etc. These chatbots were from various industries but were limited to customer service chatbots (e-commerce, online food delivery, online travel, and reservations).

3.1. Qualitative instrument

The FGD comprised six participants, two females, and four males labeled as P1 to P6. A purposive sampling strategy was used to recruit the participants. All participants were current users of customer service chatbots. The participants were between 25 and 42 years old (3 Gen-Z and 3 Millennials), with an average age of 29. The FGD lasted for an hour. Before recruitment, the participants filled out a small form about their demographics and encounters with customer service chatbots. An FGD guide was prepared comprising open-ended questions like motivation to explore chatbots, followed by their knowledge of chatbots. The subsequent questions related to their opinion on the benefits of chatbots, their interface, interactivity, user-friendliness, usefulness, the impact of peers and friends, trust and privacy, compatibility with other technologies, degree of satisfaction, intention, and usage. The FGD was audio recorded, transcribed verbatim, and subjected to content analysis using a five-step process – 1) understanding the data collected, 2) focusing the analysis, 3) categorizing the information, 4) identifying relationships

within and between constructs, and 5) conceptualizing the themes together (Powell & Renner, 2003). Four experts (Two professionals using AI, and Two AI educators) were engaged to assess the items for representativeness, comprehensiveness, and clarity. Experts were requested to provide their scores on each item independently. Items that received an agreement score of 75 percent or more across all three parameters were retained.

3.2. Survey instrument

The survey instrument comprised eight constructs measured with statements mainly drawn from established instruments validated in prior studies. These statements were adapted to fit the context of chatbot applications. First, the UTAUT model and the scale built by Venkatesh et al. (2003) were adapted to measure a user's perception regarding chatbots on various dimensions like EE, PE, FC, SI, and BI. Statements for TR were derived from Gefen et al. (2003) and Hong and Cha (2013). To measure user satisfaction (SAT), items were drawn fromargas-Avila et al. (2009). Lastly, the statements to measure adoption (AD) were drawn from Pillai and Sivathanu (2020), Melián-González et al. (2021), and Kasilingam (2020), and in line with previous studies, a Likert scale was utilized to evaluate all the statements, with responses ranging from 1, indicating "strongly disagree," to 5, representing "strongly agree".

There were three sections in the survey instrument. The first section included questions on how frequently the chatbot was used and what it is commonly used for. The second section focused on perception-based questions about factors affecting BI, SAT, and AD. The final section gathered demographic information such as gender, birth year, annual income, profession, marital status, and city. All Likert scale items were evaluated to determine normality. The kurtosis values varied from −0.813 to 0.359, and skewness values varied from −0.617 to 0.339. Across all items, skewness and kurtosis fell within the satisfactory range of −1 to +1. The results showed no notable deviation from normality (Hair et al., 2017).

3.3. Survey sample and data collection

The study participants primarily consisted of urban, college-educated student populations from Generation Z and Millennials in India having prior experience with customer support chatbots. A convenience and purposive sampling approach was utilized to gather data, a method commonly applied in past research to focus on specific groups sharing certain characteristics—in this case, a younger demographic. The criteria for selection included: (1) being within the age range of either 11 to 26 or 27 to 43, (2) having prior experience with customer support chatbots, and (3) ensuring a balanced gender representation. Combining both methods minimized the bias and ensured a large, diverse participant pool (convenience sampling) and educated, tech-savvy users (purposive sampling).

An online survey was conducted, and only respondents who satisfied the first two criteria were invited to proceed with the entire survey. A pilot survey with five respondents was conducted to identify any issues or ambiguities. It took the respondents an average of 15 min to complete the survey. Participants were recruited via personal and professional networks. Of the 2700 questionnaires distributed, 730 were completed in full, yielding a 27 % response rate. Among these, 180 were incomplete, 50 were rushed responses completed within 5 min, and 13 showed no variance in their Likert scale responses. These 243 responses were excluded, leading to a final sample of 487 for further analysis.

Table 2 presents a demographic summary of the respondents, with a gender distribution of 60 % male and 40 % female. Most respondents were Gen Z (58.7 %), while the remaining 41.3 % were Millennials. Approximately one-third of the sample was unemployed, with 85 % employed respondents working in urban areas. A large portion (95 %) of the sample were single, with the majority based in India’s Tier 1 and Tier 2 cities.

3.4. Statistical analysis

The conceptual framework for analyzing user adoption (see Fig. 2) was evaluated through Partial Least Squares Structural Equation Modeling (PLS-SEM) using SmartPLS 4.0. This method involves two steps: first, evaluating the measurement model to confirm its accuracy and consistency; and second, estimating the structural model to evaluate the hypothesized relationships. Researchers often select PLS-SEM when data do not follow a normal distribution or when sample sizes are relatively small, making it suitable for predictive and exploratory research. In the PLS application, the minimum required sample size should be at least ten times the highest number of paths leading to any

Table 2
Sample characteristics (N = 487).

Variable	Group	Frequency	Percent
Gender	Male	292	40
	Female	195	60
Age Cohort	11–26 (Gen Z)	286	58.7
	27–42 (Millennials)	201	41.3
Annual Income (INR)	Not Earning	365	74.9
	Up to 2,00,000	10	2.1
	2,00,001–5,00,000	29	6.0
	5,00,001–10,00,000	43	8.8
	10,00,001 and above	40	8.2
Profession	Service	60	12.3
	Self-employed	13	2.7
	Student	414	85.0
Marital Status	Single	464	95.3
	Married	23	4.7
City	Tier 1	270	55.4
	Tier 2	166	34.1
	Town/Village	51	10.5

Notes: Tier 1: Delhi/Mumbai/Kolkata/Chennai/Pune/Hyderabad/Bengaluru; Tier 2: Cities other than Tier 1; Town/Village: Localities other than Tier 1 & 2; INR: Indian Rupees (1 USD = INR 83 approx.).

single construct in the model. This study utilized a sample size of 487, which surpasses the lowest threshold of 60 as suggested by Hair et al. (2017).

3.5. Common method bias testing

As the sample data was gathered through self-reported responses using a single survey instrument and a uniform response method (Likert Scale) for independent and dependent variables, the possibility of common method bias was a potential concern. Harman’s single-factor test was conducted to mitigate this, which assesses if a single factor in the unrotated principal component analysis (covering all measured items) explains > 50 % of the variance (Podsakoff et al., 2003). The highest contributing factor explained 44.57 % of the variance, falling below the 50 % threshold, which indicates that common method bias is not likely to be a significant concern in this study.

4. Results

4.1. Qualitative data analysis

The following discussion summarizes the findings from the focus group discussions (FGD) through an analysis of the qualitative data gathered. The FGD revealed several key themes, shedding light on customer perceptions and the factors influencing the adoption of customer service chatbots.

4.1.1. Motivations for exploring chatbots

Participants found chatbots quicker and more convenient than traditional methods like call centers, appreciating their 24/7 availability and instant responses. Many prefer chatbots as the first point of contact for simple queries, as resolving issues with them avoids contacting a human representative. Respondents appreciated that chatbots are always accessible, unlike traditional customer service options that may have limited hours. The following excerpts highlight this notion:

“It’s quicker with chatbots; unlike calls where you wait for someone to pick up, AI-enabled chatbots respond immediately. It’s simple and convenient.” (P1, female, age 25)

“With no waiting time and 24/7 availability, chatbots are more appealing than human interactions, making them the preferred choice.” (P2, female, age 25)

4.1.2. Perceived effort needed to use chatbots

The respondents generally found chatbots easy to use and intuitive. They mentioned that handling a chatbot was straightforward and required no special skills or technical knowledge. These perceptions are captured in the following excerpts:

“I find the chatbot interface user-friendly and the interactions intuitive. The design is straightforward, making it easy to navigate and get the information I need quickly.” (P1, female, age 25)

“The chatbot interface is quite easy to use, with clear prompts and simple navigation. It’s intuitive enough; I don’t need to think too much about what to do next, which makes the experience smooth and hassle-free.” (P6, male, age 25)

4.1.3. Perceived usefulness of chatbots

Several participants appreciated the chatbot’s capabilities in terms of their efficiency (time and effort savings), particularly in scenarios requiring quick resolutions or when dealing with straightforward queries. Respondents liked that chatbots reduced the turnaround time for query resolution.

"Chatbots boost productivity by eliminating wait times and providing relevant prompts, allowing users to quickly resolve issues without the need for lengthy calls or typing out details." (P1, female, age 25).

"I used a banking chatbot to file a complaint, and it created the ticket within seconds, demonstrating impressive speed and efficiency. The issue was resolved quickly, even with human follow-up." (P4, male, age 27)

"I experienced delays with a human representative who promised a refund but didn't follow through, requiring me to raise the complaint again. With a chatbot, this could have been automated, saving time and effort by avoiding human error." (P3, male, age 29)

4.1.4. Social influence and chatbot adoption

While most respondents felt that social influence affects the adoption of generative AI technologies, its impact on chatbot adoption is less significant. While peer pressure and social trends heavily influence the use of advanced AI tools like ChatGPT, chatbot adoption is more often driven by personal convenience and necessity than social factors. This perception is reflected in the following excerpts:

"As generative AI like ChatGPT gains popularity, a fear of being left behind drives peer pressure, encouraging adoption of these technologies." (P5, male, age 42)

"I'm influenced by peers when it comes to adopting generative AI like ChatGPT, but not chatbots. Chatbot use feels more like a personal choice, and people don't discuss them as much." (P6, male, age 25)

4.1.5. Conditions that facilitate chatbot adoption

Most respondents feel confident and well-prepared to handle chatbots, indicating that they face minimal or no technical difficulties when using these tools. They express ease in navigating chatbot interfaces, effectively searching for information, and obtaining accurate and relevant responses. This suggests a high level of comfort and proficiency among users in interacting with chatbot technology.

"I feel I have the necessary technological readiness and am confident in handling chatbots, searching effectively, and obtaining useful responses." (P3, male, age 29)

"Better familiarity with chatbot features, combined with user-friendly conditions, enables me to access chatbot services smoothly and efficiently." (P1, female, age 25)

4.1.6. Customer trust with chatbots

Respondents have mixed perceptions about the security and privacy of data shared with chatbots. While some are confident that adequate protections are in place, others, especially in corporate environments, are concerned about potential risks. This caution is particularly pronounced when sensitive data is involved. Addressing these security concerns is crucial for broader adoption, especially in professional settings.

"I believe chatbots use encryption to store and remember data securely, preventing leaks. However, like most platforms, they analyze this data to influence decisions for marketing and other purposes." (P2, female, age 25)

"Trust in chatbots is conditional; participants trust them for simple, relevant queries, but trust diminishes if chatbots provide irrelevant or incorrect responses, especially with complex issues." (P3, male, age 29)

"I haven't faced any issues personally, but there's concern about privacy, especially in corporate settings. Sensitive data, like confidential sales information, shouldn't be shared with chatbots, as it could be exposed or misused. While personal use may not pose a significant risk, corporates are cautious and often restrict employees from using chatbots for important or sensitive information." (P4, male, age 27)

4.1.7. Satisfaction with chatbots

Satisfaction with chatbots is mixed among participants. While some are satisfied with their performance for basic tasks, others are frustrated by their limitations, such as difficulty handling complex queries and the robotic nature of responses. Participants tend to prefer human interaction for more complex issues or complaints, as chatbots often fall short of fully resolving these problems. The following excerpts highlight these issues:

"I find chatbots too robotic and unable to understand emotions. I would prefer using them over humans only if they could better comprehend emotions. Last week, I used a bank's chatbot to inquire about outward remittance, but it gave me information on inward remittance instead. When I tried to dig deeper, the chatbot couldn't help and suggested contacting a branch or salesperson." (P6, male, age 25)

"I believe chatbots lack empathy and cannot understand human emotions or urgency, making human interaction necessary when a customer is overwhelmed." (P2, female, age 25)

"Chatbot responses are reliable when relevant, but they can be inaccurate if the chatbot struggles to comprehend the query. When the chatbot understands the question, the information provided is accurate." (P5, male, age 42)

4.1.8. Behavioural intention toward chatbots

The respondents exhibited behavioral intention toward adopting chatbots, driven by convenience, speed, and efficiency. Many participants have already integrated chatbots into their daily routines for tasks like food delivery, booking tickets, and managing returns. This is reflected in the below two quotes:

"I think chatbots will eventually replace traditional customer service, and I'm ready to adopt them as they continue to improve in handling more complex tasks." (P2, female age 25)

"Users are increasingly inclined towards chatbots and believe that, over time, they will be preferred over human interaction, leading to a reduction in human contact." (P6, male, age 25)

4.1.9. Chatbot usage

Respondents indicate that they actively use chatbots for various tasks such as food delivery, making reservations, booking tickets, online purchases, and processing return and exchange requests. They appreciate chatbots' convenience and efficiency in completing these routine tasks. They express a willingness to continue using chatbots as illustrated in the following excerpts:

"I often use chatbots for making online reservations, booking tickets for events and shows, and processing return and exchange requests." (P4, male age 27)

"I use chatbots for food delivery to receive recommendations and suggestions based on reviews." (P5, male, 42)

4.1.10. Demographic and personality influence

Respondents believed that communication preferences vary widely across age groups. Millennials, who grew up with digital technologies, have a natural tendency to adopt them, while Gen Z, who was born into a world where technology was readily available, tends to be more open to diverse perspectives and social causes.

"Tech-savvy younger generations are more likely to adopt chatbots, particularly when the interaction is intuitive and straightforward." (P2, female, age 25)

"As someone who tends to be introverted, I find chatbots less stressful than talking to a person, making me more likely to use them." (P5, male, 42)

4.2. Measurement model assessment

The reliability of the data was evaluated by measuring internal consistency through Cronbach's Alpha (CA) and Composite Reliability (CR), with the corresponding results detailed in Table 3. All constructs showed CA and CR values above 0.70, signifying reliable measurement. Both convergent and discriminant validity assessments were conducted to validate the measurement model further.

Hair et al. (2017) have mentioned that convergent validity evaluates the degree to which the items of a multi-item construct are closely related or the extent to which a measure correlates with other measures of the same concept. This validity was tested using the following criteria: all indicator factor loadings must be at least 0.7, composite reliability (CR) for all constructs should be 0.7, and the average variance extracted (AVE) must surpass the accepted threshold of 0.5. As shown in Table 3, the factor loadings for all constructs were above 0.7, except for two indicators having loadings of 0.641 and 0.691. Both these indicators have been retained as they are theoretically important for defining the latent constructs - Behavioral Intention and Trust respectively. Removing BI5 (I will prefer chatbots over human interactions) and TR6 (Chatbots do not involve any user monitoring) may compromise the complete representation of the construct. BI5 captures a unique dimension of chatbot adoption intention, particularly among users who perceive it as an automation tool and a replacement for human interaction. Likewise, TR6 captures a unique dimension of trust, particularly the perception that chatbots operate independently without tracking or monitoring. As per Henseler et al. (2016), the cut-off of 0.7 may not always be suitable for exploratory studies; hence, values below it can be retained. Lastly, while the factor loadings are below 0.7, they are still above 0.6, meaning they still contribute to understanding the construct and are within the acceptable range for exploratory studies (Hair et al., 2021). Removing these indicators may increase AVE and composite reliability, making the model appear robust but theoretically compromised. Further, removing these indicators did not significantly improve CR or AVE values. Additionally, with CR values for all constructs above 0.7 and AVE values exceeding 0.50, the requirements for convergent validity were met.

Discriminant validity indicates how well a construct is distinguished from other constructs and was evaluated through three methods: (i) cross-loadings, (ii) the Fornell-Larcker criterion, and (iii) Heterotrait-Monotrait Ratios (HTMT). According to the cross-loadings method, must exhibit a higher loading on its associated construct compared to any other construct (Chin, 1998). As shown in Table 4, this requirement was successfully fulfilled. According to the Fornell-Larcker criterion, discriminant validity is established when the square root of the AVE for each construct exceeds the correlations with other constructs (Fornell & Larcker, 1981). In this study, the square root of the AVE for each construct (diagonal elements in Table 3) was greater than the inter-item correlations (below the diagonal). Additionally, the HTMT measures between any two reflective constructs were below the 0.9 thresholds (ranging from 0.604 to 0.863), further supporting discriminant validity.

Since all three criteria were satisfied, the measurement model was validated as reliable and accurate.

4.3. Structural model assessment

Evaluating the structural model involves estimating the causal relationships and their predictive accuracy. Before proceeding with the structural model analysis, it is essential to check for multi-collinearity issues. The Variance Inflation Factor (VIF) was calculated for the five endogenous constructs to address this. The VIF values ranged from 2.116 to 3.226 for BI, and from 1.811 to 2.818 for AD, with a value of 1.382 for SAT; all values were below the limit of 3.33, indicating no evidence of multicollinearity. Traditional collinearity assessments have often used 5 or 10 as common VIF thresholds (Hair et al., 2021). However, the choice of 3.33 is based on more recent literature, particularly in the context of PLS-SEM. Several studies in technology adoption literature, specifically AI, using PLS-SEM (Fu et al., 2024; Sharma et al., 2024; Rajan & Niranjana, 2025; Mathur et al., 2025) have used the 3.33 threshold, citing its ability to enhance model accuracy and reliability (Kock & Lynn, 2012; Kock, 2015; Sarstedt et al., 2021). According to Diamantopoulos and Siguaw (2006), researchers should adhere to the more conservative rules of thumb, such as 3.33 or 5 as suggested by Hair et al. (2021).

The next step involves assessing the structural model—Fig. 2 and Table 5 present the path analysis and hypothesis testing results. As shown in Table 5, the standardized path coefficients provide estimates and indicate the significance of the hypothesized relationships among the constructs. All the proposed hypotheses were supported except H1 (EE → BI), and H4 (FC → BI). While these two hypotheses had the desired hypothesized direction, the impact was insignificant. The path from H2: PE to BI was positive and statistically significant ($\beta_{PE \rightarrow BI} = 0.247, p = 0.000$), supporting H2. SI had a significant relationship with BI ($\beta_{SI \rightarrow BI} = 0.160, p = 0.000$), and FC had a positive and significant relationship with AD ($\beta_{FC \rightarrow AD} = 0.143, p = 0.004$) and SAT ($\beta_{FC \rightarrow SAT} = 0.374, p = 0.000$), supporting H3, H5 and H6 respectively. Trust was found to positively and significantly influence both BI ($\beta_{TR \rightarrow BI} = 0.172, p = 0.000$) and SAT ($\beta_{TR \rightarrow SAT} = 0.504, p = 0.000$), offering support to H7 and H8. Moreover, SAT had a significant impact on BI ($\beta_{SAT \rightarrow BI} = 0.346, p = 0.000$) and AD ($\beta_{SAT \rightarrow AD} = 0.250, p = 0.000$) lending support to H9 and H10. Finally, the path from BI to AD ($\beta_{BI \rightarrow AD} = 0.380, p = 0.000$) was found to be significant; therefore, H11 was supported. The results also indicate that control variables - gender, income, profession, marital status, and city did not have a confounding effect on behavioral intention to adopt service chatbots. This suggests that none of the demographic control variables significantly influenced behavioral intention. In other words, by holding constant the demographic variables, we ensure that the observed relationships are true to the constructs of interest.

The coefficient of determination (R^2), a measure of predictive power that reflects the extent to which the antecedents account for the variance in an endogenous construct within the model, is calculated by the

Table 3
Convergent and discriminant validity.

CNST	Range of FL	CA	CR	AVE	AD	BI	EE	FC	PE	SAT	SI	TR
AD	0.771–0.852	0.868	0.868	0.654	<i>0.809</i>							
BI	0.641–0.847	0.886	0.894	0.640	0.664	<i>0.800</i>						
EE	0.719–0.822	0.866	0.869	0.598	0.579	0.680	<i>0.774</i>					
FC	0.781–0.837	0.822	0.827	0.651	0.540	0.623	0.647	<i>0.807</i>				
PE	0.755–0.877	0.896	0.899	0.709	0.594	0.736	0.766	0.620	<i>0.842</i>			
SAT	0.734–0.807	0.926	0.926	0.600	0.638	0.779	0.699	0.639	0.675	<i>0.774</i>		
SI	0.791–0.866	0.857	0.857	0.701	0.558	0.684	0.572	0.663	0.690	0.633	<i>0.837</i>	
TR	0.691–0.832	0.885	0.888	0.636	0.560	0.681	0.574	0.526	0.555	0.701	0.541	<i>0.798</i>

Notes: CNST: Constructs; FL: Factor Loadings; CA: Cronbach's Alpha; CR: Composite Reliability; AVE: Average Variance Extracted; EE: Effort Expectancy; PE: Performance Expectancy; SI: Social Influence; FC: Facilitating Conditions; PR: Perceived Risk; TR: Trust; SAT: User Satisfaction; BI: Behavioral Intention; AD: Adoption; Diagonal values (italics) are the square root of AVE for each construct; non-diagonal values are inter-correlation coefficients among constructs.

Table 4
Cross-loadings.

	AD	BI	EE	FC	PE	SAT	SI	TR
AD1	0.802	0.570	0.487	0.455	0.499	0.514	0.460	0.495
AD2	0.807	0.519	0.438	0.407	0.456	0.527	0.452	0.427
AD3	0.852	0.550	0.478	0.469	0.502	0.514	0.491	0.472
AD4	0.810	0.498	0.456	0.444	0.462	0.492	0.441	0.445
AD5	0.771	0.544	0.480	0.406	0.479	0.530	0.410	0.421
BI1	0.548	0.817	0.616	0.509	0.615	0.603	0.548	0.525
BI2	0.542	0.818	0.574	0.525	0.602	0.628	0.504	0.567
BI3	0.588	0.847	0.618	0.592	0.636	0.665	0.538	0.559
BI4	0.528	0.815	0.500	0.498	0.581	0.639	0.576	0.564
BI5	0.400	0.641	0.310	0.302	0.396	0.495	0.508	0.471
BI6	0.563	0.844	0.601	0.524	0.665	0.690	0.612	0.582
EE1	0.424	0.442	0.744	0.518	0.526	0.453	0.369	0.404
EE2	0.434	0.454	0.786	0.573	0.525	0.485	0.376	0.429
EE3	0.442	0.518	0.796	0.532	0.560	0.524	0.404	0.426
EE4	0.463	0.550	0.822	0.530	0.634	0.582	0.449	0.451
EE5	0.458	0.602	0.771	0.485	0.657	0.594	0.535	0.466
EE6	0.456	0.553	0.719	0.386	0.618	0.574	0.484	0.474
FC1	0.404	0.394	0.447	0.781	0.430	0.424	0.498	0.350
FC2	0.428	0.481	0.537	0.826	0.448	0.489	0.479	0.388
FC3	0.454	0.523	0.536	0.837	0.522	0.540	0.556	0.426
FC4	0.448	0.583	0.554	0.781	0.579	0.584	0.590	0.510
PE1	0.543	0.634	0.656	0.540	0.839	0.542	0.609	0.484
PE2	0.486	0.631	0.644	0.531	0.877	0.593	0.614	0.471
PE3	0.476	0.635	0.627	0.512	0.875	0.579	0.583	0.470
PE4	0.516	0.629	0.614	0.509	0.857	0.582	0.575	0.432
PE5	0.476	0.564	0.689	0.520	0.755	0.546	0.519	0.480
SAT1	0.509	0.607	0.495	0.401	0.551	0.737	0.532	0.532
SAT2	0.474	0.575	0.566	0.529	0.495	0.773	0.480	0.536
SAT3	0.514	0.585	0.485	0.444	0.503	0.734	0.523	0.481
SAT4	0.532	0.609	0.537	0.497	0.550	0.807	0.510	0.553
SAT5	0.486	0.609	0.531	0.489	0.507	0.783	0.481	0.616
SAT6	0.519	0.613	0.530	0.517	0.529	0.799	0.482	0.584
SAT7	0.471	0.613	0.553	0.458	0.544	0.791	0.501	0.526
SAT8	0.519	0.651	0.607	0.548	0.544	0.805	0.518	0.576
SAT9	0.456	0.608	0.595	0.545	0.518	0.767	0.455	0.543
SAT10	0.453	0.555	0.511	0.511	0.486	0.743	0.422	0.466
SI1	0.433	0.560	0.489	0.546	0.577	0.485	0.850	0.432
SI2	0.442	0.579	0.437	0.523	0.554	0.513	0.866	0.436
SI3	0.514	0.581	0.538	0.593	0.586	0.578	0.791	0.501
SI4	0.477	0.567	0.450	0.555	0.591	0.542	0.840	0.441
TR1	0.465	0.588	0.523	0.440	0.524	0.591	0.460	0.802
TR2	0.484	0.530	0.487	0.490	0.424	0.575	0.405	0.825
TR3	0.455	0.588	0.518	0.449	0.464	0.565	0.453	0.831
TR4	0.425	0.545	0.466	0.442	0.410	0.573	0.396	0.832
TR5	0.441	0.532	0.410	0.400	0.472	0.545	0.476	0.796
TR6	0.405	0.467	0.323	0.278	0.347	0.499	0.399	0.691

Notes: EE: Effort Expectancy; PE: Performance Expectancy; SI: Social Influence; FC: Facilitating Conditions; PR: Perceived Risk; TR: Trust; SAT: User Satisfaction; BI: Behavioral Intention; AD: Adoption; All items with loadings >0.6 were retained.

SmartPLS tool. The proposed model explains 72.8 % of the variance in BI, 48.9 % in AD, and 59.2 % in SAT. R^2 values of 75 %, 50 %, and 25 % are generally classified as substantial, moderate, and weak, respectively. This indicates that all endogenous constructs in this study possess a moderate level of predictive power.

The predictive relevance (Q^2) and effect size (f^2) were also evaluated, as shown in Table 6. The cross-validated predictive relevance of the structural model was calculated using the Stone-Geisser Q^2 value with an omission distance of 7. A higher Q^2 value suggests better predictive accuracy. For this study, Q^2 values were found to be 0.586, 0.663, and 0.426 for SAT, BI, and AD, respectively. Since all Q^2 values are above zero, this confirms that the model has predictive relevance and is well-constructed.

Subsequently, for each endogenous construct (refer to Table 6) effect size f^2 was computed. According to Cohen (1988), f^2 values of 0.02, 0.15, and 0.35 correspond to small, medium, and large effects, respectively. For the endogenous construct BI, the constructs PE, SAT, SI, and TR demonstrated a small effect size ($0.02 \leq f^2 \leq 0.15$), while EE and FC had negligible effect sizes, as their f^2 values were <0.02. Regarding AD, all constructs FC, SAT, and BI showed small effects. For the construct SAT, TR displayed a large effect size, while FC showed a medium effect.

4.4. Overall model fit

Goodness of Fit (GoF) is used as a general indicator of model adequacy, which is computed as a square root of the product of the average R square and geometric mean of the average communality (AVE). The computed GoF value of 0.62 is considered large as it exceeds the cut-off value of 0.36. This implies a good model fit. Additionally, to ensure that the model adequately accounts for the data, two approximate fit indices (SRMR, and rms Theta) are computed and their values are compared to a threshold (e.g., SRMR < 0.08). The value of SRMR was found to be 0.054, which is below the thresholds, suggesting that the underlying model demonstrates a strong fit. The value of rms Theta is found to be 0.112.

4.5. Multigroup analysis: Gen Z and Millennial users

Before conducting the multi-group analysis using PLS-MGA, an independent sample t -test was conducted between the two groups: Gen Z ($n = 286$) and Millennials ($n = 201$) from the urban, student sample. The mean values for all constructs were computed and examined for significant distinctions within this demographic. The results indicate

Table 5
Hypotheses testing results of the structural model.

Hypotheses	Relationship	Path Coefficient	t-value	p-values	Supported
H1	EE -> BI	0.044	0.891	0.373 ^{n.s.}	No
H2	PE -> BI	0.247	5.256	0.000*	Yes
H3	SI -> BI	0.160	4.315	0.000*	Yes
H4	FC -> BI	0.010	0.234	0.815 ^{n.s.}	No
H5	FC -> AD	0.143	2.911	0.004*	Yes
H6	FC -> SAT	0.374	10.138	0.000*	Yes
H7	TR -> BI	0.172	4.190	0.000*	Yes
H8	TR -> SAT	0.504	13.607	0.000*	Yes
H9	SAT -> BI	0.346	7.205	0.000*	Yes
H10	SAT -> AD	0.250	3.701	0.000*	Yes
H11	BI -> AD	0.380	6.068	0.000*	Yes
Control variables					
Gender	Gender -> BI	-0.017	0.346	0.729 ^{n.s.}	
Income	Inc -> BI	-0.054	1.754	0.079 ^{n.s.}	
Profession	Prof -> BI	-0.032	0.878	0.380 ^{n.s.}	
Marital Status	MS -> BI	0.313	1.847	0.065 ^{n.s.}	
City	City -> BI	-0.015	0.681	0.496 ^{n.s.}	

Notes: EE: Effort Expectancy; PE: Performance Expectancy; SI: Social Influence; FC: Facilitating Conditions; PR: Perceived Risk; TR: Trust; SAT: User Satisfaction; BI: Behavioral Intention; AD: Adoption; Control variables include Gender, Income (Inc), Profession (Prof), Marital Status (MS), and City; * $p < 0.01$; n.s.: not significant.

Table 6
 f^2 and q^2 values for the endogenous constructs BI and SAT.

Behavioural Intention (BI)	Path	R ²	f ²
All constructs included		0.728	
EE excluded	EE->BI		0.002
FC excluded	FC->BI		0.001
PE excluded	PE->BI		0.071
SAT excluded	SAT->BI		0.144
SI excluded	SI->BI		0.038
TR excluded	TR->BI		0.051
Adoption (AD)	Path	R²	f²
All constructs included		0.489	
FC excluded	FC->AD		0.022
SAT excluded	SAT->AD		0.043
BI excluded	BI->AD		0.104
User Satisfaction (SAT)	Path	R²	f²
All constructs included		0.592	
FC excluded	FC -> SAT		0.247
TR excluded	TR -> SAT		0.451

significant differences between the two user groups within this study sample. Compared to Gen-Z, Millennials in the sample reported significantly higher values for all constructs: EE ($M_m=3.438$; $M_z=3.195$; $p < 0.01$), PE ($M_m=3.234$; $M_z=2.984$; $p < 0.01$), SI ($M_m=3.054$; $M_z=2.892$; $p < 0.05$), FC ($M_m=3.482$; $M_z=3.182$; $p < 0.01$), TR ($M_m=3.140$; $M_z=2.980$; $p < 0.05$), SAT ($M_m=3.250$; $M_z=3.078$; $p < 0.01$), BI ($M_m=3.169$; $M_z=2.988$; $p < 0.05$) and AD ($M_m=3.084$; $M_z=2.909$; $p < 0.05$).

As the subsequent step, a PLS-MGA was conducted utilizing the bias-corrected and accelerated (BCa) bootstrap method, which was recognized as the most stable approach. Before running this method, the measurement invariance of composite models (MICOM) permutation procedure was applied to access measurement variance in the composite model (Henseler et al., 2016). It is a three-step procedure to check for configural invariance, compositional invariance, and the equality of composite mean values and variances (Henseler et al., 2016). Each step

is a precondition for assessing the subsequent step. In Step 1, since the same set of indicators consistently relate to the same constructs across groups, with identical signs on their loadings, configural invariance is automatically established. In Step 2, to check for compositional invariance, a permutation test was used to check if the original correlation is greater than the five-percent quantile. Compositional variance was established since all the original correlation values exceeded the five percent quantile, fulfilling the primary condition of partial measurement invariance. Finally, Step 3 was examined in two parts (Step 3a and Step 3b) to establish the equality of the composite mean values and variances for males and females. In Step 3a, the equality of mean values is not established since the mean differences for each construct do not fall within the lower (2.5 %) and upper (97.5 %) interval. Further, the variance differences for males and females lie in the confidence intervals only for AD, EE, FC, PE, SAT, and TR. The results of MICOM analysis support configural and compositional invariance but do not equal mean values and variance. Thus, partial measurement invariance is confirmed allowing for the comparison of standardized coefficients of the structural model across different groups. The results are summarized in Table 7.

Further, PLS-MGA was done to determine whether the differences in path coefficients between Gen Z and Millennial users differed significantly. We did not find statistically significant moderated effects of age cohort in any relationship, except for the path from SAT to BI. As evident from Table 8, the path coefficient difference for SAT to BI is significantly higher for Gen Z (p -value=0.011), signifying that SAT has a more substantial influence on BI for Gen Z than millennials. For the other relationships, the difference was insignificant.

5. Discussion

The primary aim of this research was to validate a model of integrating TR with the UTAUT framework and satisfaction to explain the utilization of chatbots in the Indian context. Although there have been prior attempts to integrate UTAUT with context-specific factors, the proposed integrated model in this study has confirmed the proposed hypotheses besides predicting chatbot adoption behaviors. Our empirical analysis supports all the proposed hypotheses, except for H1: EE -> BI, and H4: FC -> BI. The lack of support for H1 (EE -> BI) suggests that users may not perceive effort expectancy as a significant barrier to chatbot adoption, possibly due to the intuitive design of modern chatbots. Similarly, the lack of support for H4 (FC-> BI) indicates that users may not perceive facilitating conditions as a significant barrier to chatbot adoption due to the existence of requisite knowledge, resources, and support. Further, the age cohort was found to moderate one relationship (SAT -> BI) in the proposed model, with the influence stronger for Gen Z than millennials. The significant findings are summarized and discussed below.

- The results of hypotheses about BI (H2, H3, H7, and H9) showed that PE, SI, TR, and SAT could lead to a positive intention towards chatbot adoption.
- Satisfaction appears to be the most influential driver behind behavioral intention. Our study supports the findings of Ashfaq et al. (2020), who found that user satisfaction determines behavioral intention (Hsiao & Chen, 2022; Hari et al., 2022). This finding aligns with recent research by Bhatnagr and Rajesh (2024), suggesting that when users feel satisfied with their chatbot experience, they are more likely to use chatbots.
- Next, PE and TR emerged as essential determinants, while users recognized SI as having a significant impact on chatbot intention. Various recent studies have shown that PE has a more significant effect on the intention to use chatbot services (Balakrishnan et al., 2022; Silva et al., 2023; ElSayed & Mamdouh, 2024; Bhatnagr & Rajesh, 2024).

Table 7

Results of MICOM Analysis for assessing measurement invariance.

Step 1		Step 2			Step 3a			Step 3b			
Cons-struct	Conf. invar.	Original correl.	5 % quantile	Comp. invar.	Mean diff.	CI (mean)	Equal mean	Var. diff.	CI (variance)	Equal (variance)	Measurement invariance
AD	Yes	0.999	0.999	Yes	−0.201	−0.171,0.154	No	0.012	−0.229,0.241	Yes	Partial
BI	Yes	0.999	0.999	Yes	−0.228	−0.171,0.169	No	−0.342	−0.274,0.263	No	Partial
EE	Yes	0.997	0.997	Yes	−0.309	−0.191,0.174	No	0.118	−0.292,0.261	Yes	Partial
FC	Yes	0.999	0.997	Yes	−0.375	−0.187,0.174	No	0.130	−0.278,0.297	Yes	Partial
PE	Yes	1.000	0.999	Yes	−0.288	−0.182,0.168	No	−0.093	−0.229,0.245	Yes	Partial
SAT	Yes	0.999	0.999	Yes	−0.242	−0.174,0.176	No	−0.233	−0.317,0.315	Yes	Partial
SI	Yes	0.998	0.998	Yes	−0.208	−0.187,0.167	No	−0.512	−0.282,0.292	No	Partial
TR	Yes	1.000	0.999	Yes	−0.209	−0.182,0.175	No	−0.144	−0.275,0.255	Yes	Partial

Table 8

Multigroup analysis: Age (Gen Z vs Millennials) as a moderator.

Hypotheses	Paths	Path coefficients (t-value)		Path coefficients differences	Supported
		Gen Z	Millennials		
H12a	EE ->	0.051 ^{n.s.}	0.015 ^{n.s.}	0.036 ^{n.s.}	No
	BI	(0.788)	(0.219)		
H12b	PE ->	0.205*	0.352*	−0.148 ^{n.s.}	No
	BI	(3.417)	(4.73)		
H12c	SI ->	0.122**	0.197*	−0.074 ^{n.s.}	No
	BI	(2.321)	(3.794)		
H12d	FC ->	0.030 ^{n.s.}	0.036 ^{n.s.}	−0.006 ^{n.s.}	No
	BI	(0.570)	(0.567)		
H12e	FC ->	0.147***	0.123***	0.023 ^{n.s.}	No
	AD	(2.185)	(1.722)		
H12f	FC ->	0.383*	0.358*	0.026 ^{n.s.}	No
	SAT	(8.280)	(6.045)		
H12g	TR ->	0.146**	0.207*	−0.061 ^{n.s.}	No
	BI	(2.494)	(3.803)		
H12h	TR ->	0.518*	0.491*	0.027 ^{n.s.}	No
	SAT	(11.225)	(8.355)		
H12i	SAT	0.436*	0.192*	0.243**	Yes
	-> BI	(6.739)	(2.699)		
H12j	SAT	0.273*	0.235*	0.038 ^{n.s.}	No
	-> AD	(2.644)	(2.707)		
H12k	BI ->	0.350*	0.429*	−0.079 ^{n.s.}	No
	AD	(3.627)	(5.468)		

Notes: Gen Z: Age 11 to 26 years; Millennials: Age 27 to 42 years; EE: Effort Expectancy; PE: Performance Expectancy; SI: Social Influence; FC: Facilitating Conditions; PR: Perceived Risk; TR: Trust; SAT: User Satisfaction; BI: Behavioral Intention; AD: Adoption; Values in bold are significant.

* indicates significance at 1 percent.

** indicates significance at 5 percent.

*** indicates significance at 10 percent; n.s.- not significant.

- In the review of our findings, we found a strong influence of TR on behavioral intention. Trust is developed when the user shows confidence in the perceived security of the platform. This finding is in line with recent studies (Mostafa & Kasamani, 2022; ElSayad & Mamdouh, 2024) which found that initial trust in chatbots drives customers to revisit the platform and encourage others like friends and family, to consider that chatbot platform. Lei et al. (2021) have mentioned that customers with greater trust are more inclined to reuse chatbots. Przegalinska et al. (2019) mention that trust is a central construct for successful human-computer interaction. SI also explains chatbot usage intention (Melián-González et al., 2021; Silva et al., 2023). The perspective of these social referents (people whom the user knows in general, and people who influence the user) should also be considered (Melián-González, 2021).
- Aligned with previous studies, the current research found that TR (H8) demonstrated a notable and positive effect on SAT with chatbot services. According to Przegalinska et al. (2019), trust is crucial as users are skeptical about sharing personal information and is particularly relevant in domains such as financial care, healthcare, and other fields that require sensitive data. Past studies (Hsiao and

Chen, 2021; Eren, 2021) have shown that confidence in chatbot services positively impacts customer satisfaction.

- The results of this research indicate that FC (H6) significantly influences SAT. This finding supports the literature. Prior computer and technology experience (Karjaluoto et al., 2002) along with the presence and cost-effectiveness of smartphones and internet access, chatbot knowledge, security, and privacy (Chawla & Joshi, 2019) determine satisfaction.
- The association between the individual constructs FC, SAT, BI, and AD proposed in H5, H10, and H11 was supported. Users believe that the existence of FC supports the adoption of chatbot services. This result is supported by Yang (2010) who found that FC facilitates the adoption of technology. Likewise, SAT and BI were found to be the most crucial variables that directly influence chatbot adoption. SAT and BI generally go hand in hand. When SAT toward chatbots increases, the behavioral intention and the associated adoption significantly increase. Moreover, the findings of this research corroborate the work of Ashfaq et al. (2020), Hsia and Chen (2021), and Hari et al. (2022) who suggested that SAT is a significant determinant of behavioral intention to adopt chatbot applications. Likewise, the findings of this research were aligned with the work of Pillai and Sivathanu (2020) who in the context of tourism chatbots found that actual usage of AI-powered chatbots is a function of adoption intention.
- Concerning the moderating effect of age cohort between SAT and BI, the difference between Gen Z and millennials is found to be significantly higher for Gen Z. The individual influence of PE, SI, FC, and TR on BI was higher for millennials as compared to Gen Z although insignificant. On the contrary, the impact of EE on BI was higher for Gen Z yet insignificant. While previous studies (Ashfaq et al., 2020; Bhatnagar & Rajesh, 2024) have established the importance of satisfaction in technology adoption, our research uniquely highlights the moderating role of age cohorts, indicating that Gen Z's satisfaction levels are stronger and more stable in influencing their behavioral intentions compared to millennials.

6. Theoretical and managerial implications

This investigation makes several theoretical contributions to chatbots, human-computer interactions, and user behavior. This research adds to the further development of this theory regarding the acceptance and utilization of chatbots in various ways by exploring actual chatbot users' satisfaction, intention, and adoption from the perspective of the extended UTAUT model. The current study further extends the available research by incorporating the trust dimension into the model and exploring the antecedents and consequences of trust. Further, the proposed model emphasizes the importance of meeting user expectations and experiences by integrating user satisfaction. More importantly, the model provides a holistic outlook on how these relationships differ across age cohorts (Gen Z and millennials). The extended version of the UTAUT shows good validity and exploratory power.

This study highlights several managerial implications.

First, users' perception of a chatbot's capability to boost performance and efficiency plays a crucial role in their willingness to adopt it. Chatbots are often used to address inquiries, clarify uncertainties, conduct searches, and deliver pertinent information. Companies providing chatbot services should ensure that users receive quick, clear responses that are easy to interpret and guide them toward follow-up questions if needed. Chatbots should also be regularly trained and updated to evolve into more sophisticated conversational agents. Additionally, managing escalation processes effectively is essential, so unresolved issues are redirected appropriately without overwhelming users with repeated queries. Implementing these strategies can assist technology experts in enhancing user satisfaction and chatbot effectiveness.

Second, social influence (SI) depends on the context in which a chatbot is utilized, whether for personal or organizational purposes. For personal chatbot use, friends and family can act as influencers, while in a professional environment, colleagues and management can affect technology adoption. Given that this study focuses on an urban, student sample from Gen Z and Millennials, companies targeting a similar demographic could leverage social media and mobile advertisements to enhance awareness of chatbot functionalities. Since young, urban students are highly active on digital platforms, these channels could drive engagement and adoption within this cohort.

Third, trust (TR) and facilitating conditions (FC) are essential for user satisfaction. For chatbot adoption to expand across diverse user groups, including those with low literacy levels and limited access to technology, chatbot providers should design services compatible with basic smartphones and reliable internet connections. However, given the limited rural representation in this study, further research is needed to explore chatbot adoption factors in rural communities. Additionally, supporting local languages in chatbot interactions is vital, as users expect relevant, helpful suggestions reinforcing trust. High-quality service and accurate information delivery, particularly in initial interactions, are crucial to building user confidence. Trust can also be strengthened through professional and reliable interactions, advice, privacy assurance, and transparent data handling. Awareness sessions to educate users about data collection, storage, and security practices could further instill confidence. Furthermore, incentives such as coupons or cashback may also encourage users to explore the platform's security.

The stronger influence of satisfaction on behavioral intention for Gen Z suggests that chatbot providers should prioritize user experience design tailored to this demographic, potentially incorporating gamification elements to enhance engagement. Gamification elements could include a progress bar to track the progress of customer queries, accumulating points or badges for getting their query resolved quickly and efficiently, award points for giving feedback or participating in polls, surveys, etc. Besides fun, by incorporating the gamified elements, service providers can get valuable feedback and insights which can be used to strengthen customer service, satisfaction, and loyalty.

In conclusion, this research suggests that chatbot providers should enhance performance expectancy, social influence, trust, and facilitating conditions to boost user satisfaction, intention, and chatbot utilization.

7. Limitations and future research avenues

While this study provides valuable insights, its limitations open avenues for future research.

The primary limitation of this study is the generalizability, as the sample used is comprised primarily of young urban college-educated students, which may not represent the broader population of Gen Z and Millennial users. The sample was selected based on the [Statista \(2024a\)](#) report wherein the age groups 18–24, 25–34, and 35–44 were most likely to use chatbots. As per another report on global use of ChatGPT by [Statista \(2024b\)](#), these groups were the most likely to use ChatGPT. While industry reports suggest that the 18–44 age group accounts for 60 to 70 percent of global chatbot users, the findings reported

in the study are specific to an urban, student population of Gen Z and Millennials in India and may not be directly generalizable to broader chatbot user demographics. Although the study's insights on chatbot adoption and user behavior may apply to other AI systems—particularly those emphasizing ease of use, complexity, usefulness, and interactivity—their relevance to other age groups and non-student populations requires further investigation.

Secondly, this study used convenience and purposive sampling, selecting participants from Tier 1, Tier 2, and small towns in India, which may limit the findings' applicability to other cultural contexts. Non-probability sampling methods may introduce sampling bias, and thus future studies could employ probability sampling techniques. Most respondents were from Tier 1 and Tier 2 cities, resulting in lower representation from small towns. The predominance of respondents from Tier 1 and Tier 2 cities may skew the findings, suggesting that future research should aim for a more balanced demographic representation to enhance the generalizability of the results. Although this study includes few respondents from small towns and villages (10.5 % of the sample), most are from urban areas. Thus, the findings and recommendations primarily reflect urban chatbot adoption patterns, where literacy levels and technology infrastructure are superior to their rural counterparts. Future research should examine adoption factors in rural settings to provide more representative insights. In addition, cross-cultural comparisons can be carried out with chatbot users from other developing Asian economies like Indonesia, the Philippines, Vietnam, Malaysia, Bangladesh, etc. where the adoption of emerging technologies like chatbots is on the rise.

Thirdly, this research focused on four constructs from the UTAUT model. As the UTAUT2 model introduces additional constructs such as hedonic motivation, habit, and price value, future research incorporating these factors could offer a deeper understanding of chatbot usage intentions.

Fourth, this study treated chatbot applications as a single, broad category. Future research could examine how the hypothesized relationships and the model's predictive power vary across specific chatbot applications, such as those used in different industries like online shopping, banking, healthcare, and tourism where chatbots are increasingly deployed to enhance user interaction. Further, the chatbot design, features, and capabilities may differ across industries, and thus the trust and satisfaction levels. In this study, the commonality between chatbots is that all are used for customer service. Therefore, conducting future studies examining how trust and satisfaction differ between different industries to generate industry-specific recommendations would be worthwhile.

Fifth, although we collected demographic data, including gender, income, and education, prior studies have explored the moderating effects of these variables. Future research could extend this work by examining these demographic influences on chatbot adoption.

Sixth, the study was cross-sectional; therefore, a longitudinal study may be carried out to capture the changing user preferences with the evolution of chatbot technologies.

Lastly, the study mainly examined young users who are quite different from other cohorts in terms of chatbot perceptions, experiences, and use of chatbots. Future research may be carried out to segment the users and profile them based on their socio-economic and demographic characteristics.

8. Conclusion

The study explored the determinants influencing the adoption of chatbot applications primarily among urban, college-educated users within Generation Z and Millennials by extending the UTAUT model with trust and satisfaction dimensions.

The qualitative and quantitative findings suggest that the identified antecedents play an important role in shaping the intention and adoption of chatbots. These factors were found to influence users' willingness

to engage with chatbot technology within customer service contexts, though the degree of influence may vary based on user demographics. Additionally, the findings highlight the different perspectives and technology preferences between Gen Z and Millennials within the study sample. Among the urban, student sample in this study, Gen Z users, as digital natives, demonstrated a higher comfort level with chatbot adoption, particularly valuing instant responses and seamless integration into their digital interactions. Conversely, Millennials in the sample, while also tech-savvy, exhibited a stronger preference for human interaction in certain situations, particularly when dealing with complex inquiries or seeking personalized service.

The empirical findings provide insights into how performance expectancy, social influence, trust, and user satisfaction influence behavioral intentions to use chatbots among urban, student populations from Gen Z and Millennials in India. Within this demographic, trust and facilitating conditions were particularly influential in enhancing user satisfaction, which in turn, positively impacts adoption. The research highlights the important role of trust in chatbot adoption. Given the inherent risks and concerns about privacy in digital interactions, trust emerges as a critical factor that can significantly influence user satisfaction and behavioral intention. The study suggests that chatbot providers targeting similar user demographics should prioritize building and maintaining trust through transparent communication, robust security measures, and ensuring a high-quality user experience. Facilitating conditions, such as the availability of reliable technical infrastructure and user support, also play a crucial role in user satisfaction and should not be overlooked. However, the impact of these factors may vary across user groups, industries, or cultural settings, which necessitates future research in different contexts.

The generational analysis highlights distinct differences between urban student populations from Gen Z and Millennials within the study sample. While both cohorts show a willingness to adopt chatbots, Gen Z's skepticism about chatbots' efficacy suggests a need for more targeted strategies to address their unique concerns and expectations. Millennials, having grown up with evolving technology, generally reported higher trust and satisfaction levels, indicating a smoother adoption process.

From a practical standpoint, this study provides context-specific insights for organizations seeking to deploy chatbots among similar user groups. By focusing on enhancing trust and satisfaction, organizations may increase chatbot acceptance and usage. This involves ensuring reliable performance, leveraging social influence, and tailoring strategies to address different generational cohorts' specific needs and preferences.

This study extends the UTAUT model by incorporating trust and satisfaction to better understand chatbot adoption among urban, student populations from Gen Z and Millennials. The findings suggest that trust and satisfaction play important roles in influencing chatbot adoption, providing context-specific insights for both academic researchers and industry professionals in chatbot development and deployment. While this study provided a framework for examining chatbot adoption, future research is needed to examine these relationships across various user groups, industry settings, and evolving AI advancements to enhance generalizability.

In summary, as chatbot technology continues to evolve, understanding the context-specific factors influencing user adoption, particularly trust and satisfaction, will be essential for developers and organizations aiming to optimize chatbot deployment among urban, student populations from Gen Z and Millennials in India. Transparency in data usage and seamless transition to human agents for complex queries can improve trust and satisfaction. Furthermore, as examined in this study, chatbots should be designed to deliver personalized experiences tailored to the preferences observed within these specific generational cohorts to drive adoption among urban, educated students from Gen Z and Millennials. Additionally, incorporating user-centric design elements such as transparency in data usage, gamification, visually

engaging interfaces, and real-time feedback mechanisms may enhance engagement levels.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author used [Grammarly and ChatGPT] to [check for sentence completion and language improvement]. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

CRediT authorship contribution statement

Himanshu Joshi: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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